

Analyzing E-Learning Platform Purchases Using MySQL

Project Overview

This project analyzes purchase data from an e-learning platform using MySQL. A relational database was designed with three tables — learners, courses, and purchases — to study learner behavior, sales performance, and course popularity using SQL joins, aggregations, and analytical queries.

Key Insights from Analysis

1. Learner Spending Analysis

Using aggregate queries, total spending by each learner was calculated based on course quantity and unit price. This helped identify high-value learners and understand purchasing behavior across countries.

2. Top Performing Courses

The top 3 most purchased courses were identified based on total quantity sold. These courses showed strong learner demand and indicate which offerings are driving engagement and sale

3. Category Revenue Performance

Revenue was analyzed by course category, along with the number of unique learners in each category. This revealed which categories generate the highest revenue and attract the most learners.

4. Multi-Category Learner Behavior

Some learners purchased courses from more than one category, indicating broader interests and cross-category engagement opportunities.

5. Unpurchased Course Identification

One course was found with no purchases, highlighting a possible low-performing course that may need improvement or stronger promotion.

6. Revenue by Country

Revenue was analyzed across countries to identify which regions contributed the highest sales. This helps understand geographic demand and can support region-specific marketing strategies.

7. Average Spending Per Learner

Average purchase spending per learner was calculated to evaluate transaction behavior and identify higher-value learners.

Recommendations

- Focus marketing efforts on top-performing courses and high-revenue categories.
- Promote underperforming or unpurchased courses through discounts or improved content positioning.
- Use cross-selling strategies for learners interested in multiple categories.
- Target high-spending learners with personalized offers or advanced courses.
- Expand popular course categories to increase revenue potential.
- Focus promotions in high-revenue countries to maximize growth.
- Use learner spending patterns to create personalized offers and upsell advanced courses.

Skills Applied

- Database design and schema creation
- SQL joins (INNER, LEFT, RIGHT)
- Aggregation using GROUP BY, HAVING, ORDER BY
- Revenue and spending calculations
- Business insight generation through SQL analysis

Conclusion

This project demonstrates how MySQL can be used to transform transactional purchase data into actionable business insights. The analysis supports decision-making for sales optimization, learner engagement, and course strategy.